Data & AI

15. How an LLM Answers

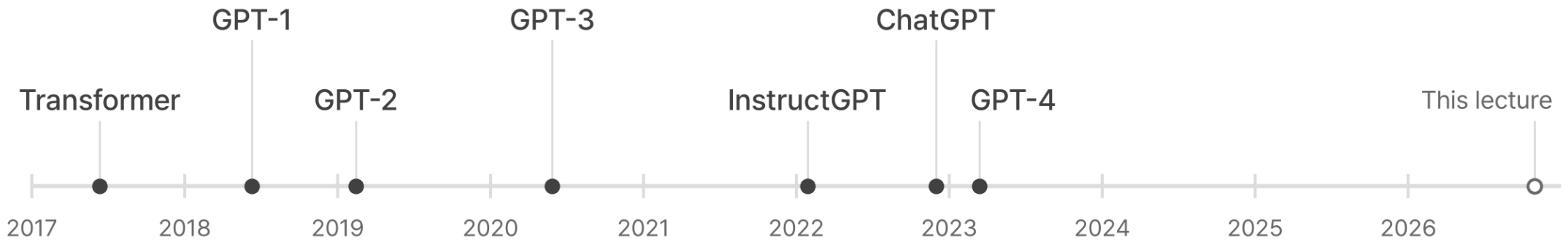
Dasaem Jeong · Art & Technology, Sogang University

Assignment 2 Mid-point

jdasam.github.io/mas1004-leaderboard

- The leaderboard this morning
- A few of you: one change you made, and what it did to your validation score
- Tuesday 10 November, 22:00: Assignment 2 due

Last Time



- Every model from GPT-1 to GPT-4 did one thing: predict the next word
- What changed: the training text, the size, and human feedback

Today

- Where an answer comes from: the parameters, the context, and tools
- Reasoning models: writing the steps before the answer

Where an Answer Comes From

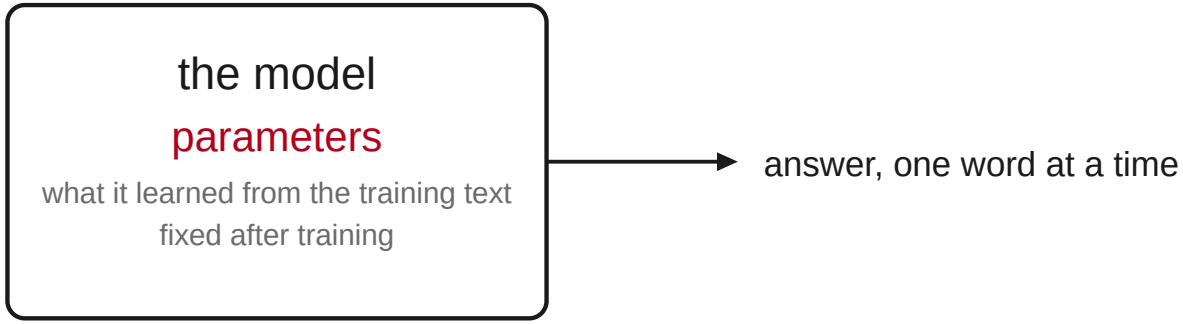
Two Questions

> At what temperature does water boil?

> Who hit the game-winning hit for SK against Hanwha in Daejeon on 7 June 2009?

- Both facts are old and public. Try each with web search off, then on.

Answers from the Parameters



- Nothing is looked up: the answer is whatever the training text made likely

Closed-Book Question Answering

Trivia questions, with nothing but the question as input

TriviaQA questions, in the format GPT-3 was tested with

> Q: 'Nude Descending A Staircase' is perhaps the most famous painting by which 20th century artist?
> A: Marcel Duchamp

> Q: What is solid CO2 commonly called?
> A: dry ice

> Q: Who wrote the novel The Eagle Has Landed?
> A: Jack Higgins

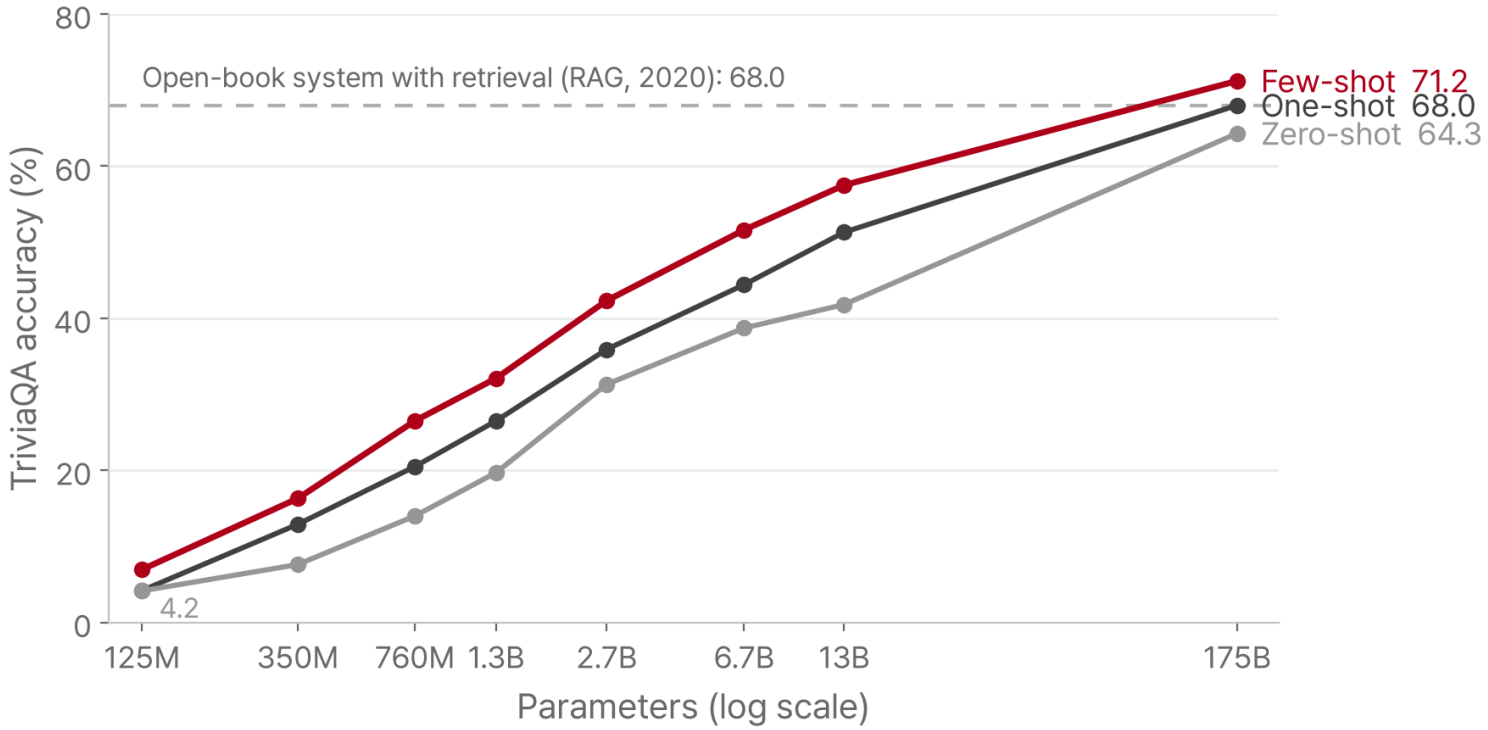
> Q: The Dodecanese Campaign of WWII that was an attempt by the Allied forces to capture islands in the Aegean Sea was the inspiration for which acclaimed 1961 commando film?
> A: The Guns of Navarone

- No document, no search: the answer has to be in the parameters already

Joshi et al. (2017), TriviaQA · Brown et al. (2020), Figure G.34 · "closed-book": Roberts, Raffel, Shazeer (2020)

TriviaQA Accuracy and Model Size

Brown et al. (2020), Language Models are Few-Shot Learners: eight sizes of GPT-3



- 125M parameters: 4 out of 100 right. 175B parameters: 64 out of 100

Brown et al. (2020), Table H.1 (dev set) and Table 3.3 · T5 sizes in Roberts et al. (2020), Table 1

Small and Large Models on SimpleQA

OpenAI (2024), SimpleQA: 4,326 short fact questions, one indisputable answer each

> Who received the IEEE Frank Rosenblatt Award in 2010? Michio Sugeno

> What day, month, and year was Carrie Underwood's album "Cry Pretty" certified Gold by the RIAA?
> October 23, 2018

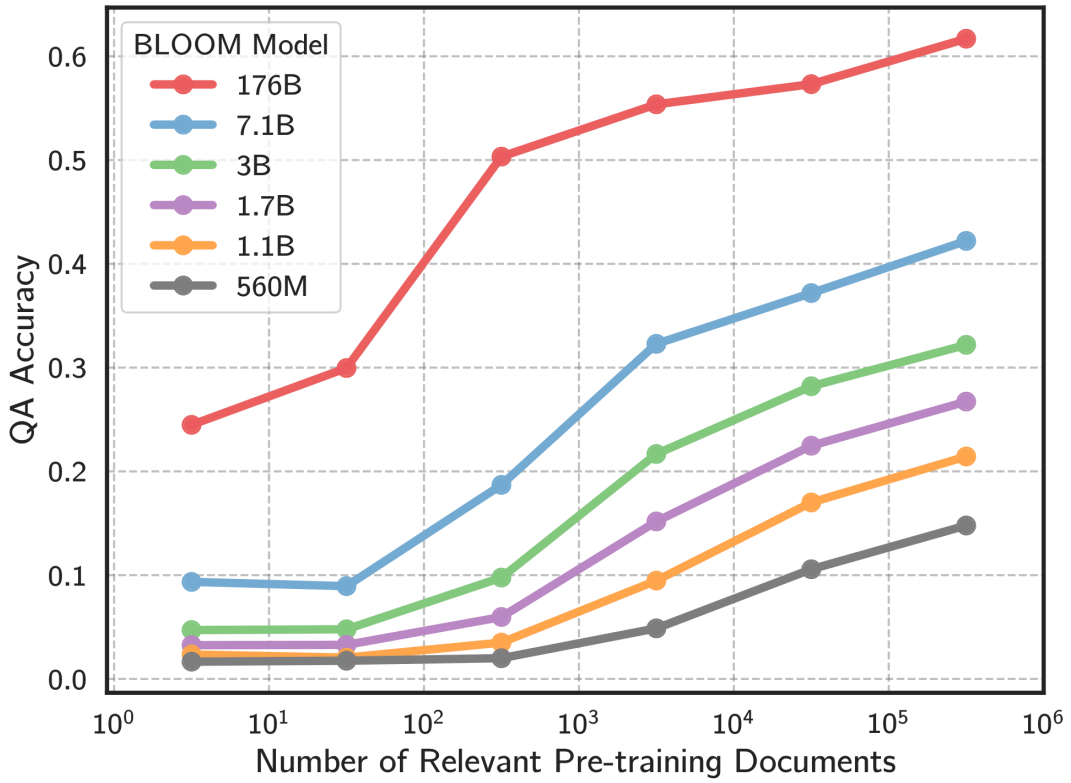
model	correct	not attempted	incorrect
Claude 3 Haiku	5.1%	75.3%	19.6%
Claude 3 Opus	23.5%	39.6%	36.9%
GPT-4o mini	8.6%	0.9%	90.5%
GPT-4o	38.2%	1.0%	60.8%

- The small model of each family gets far fewer right. One abstains, one guesses

Wei et al. (2024), Measuring short-form factuality in large language models, Table 1 and Table 3

Long-Tail Knowledge

Kandpal et al. (2023), Large Language Models Struggle to Learn Long-Tail Knowledge, ICML 2023

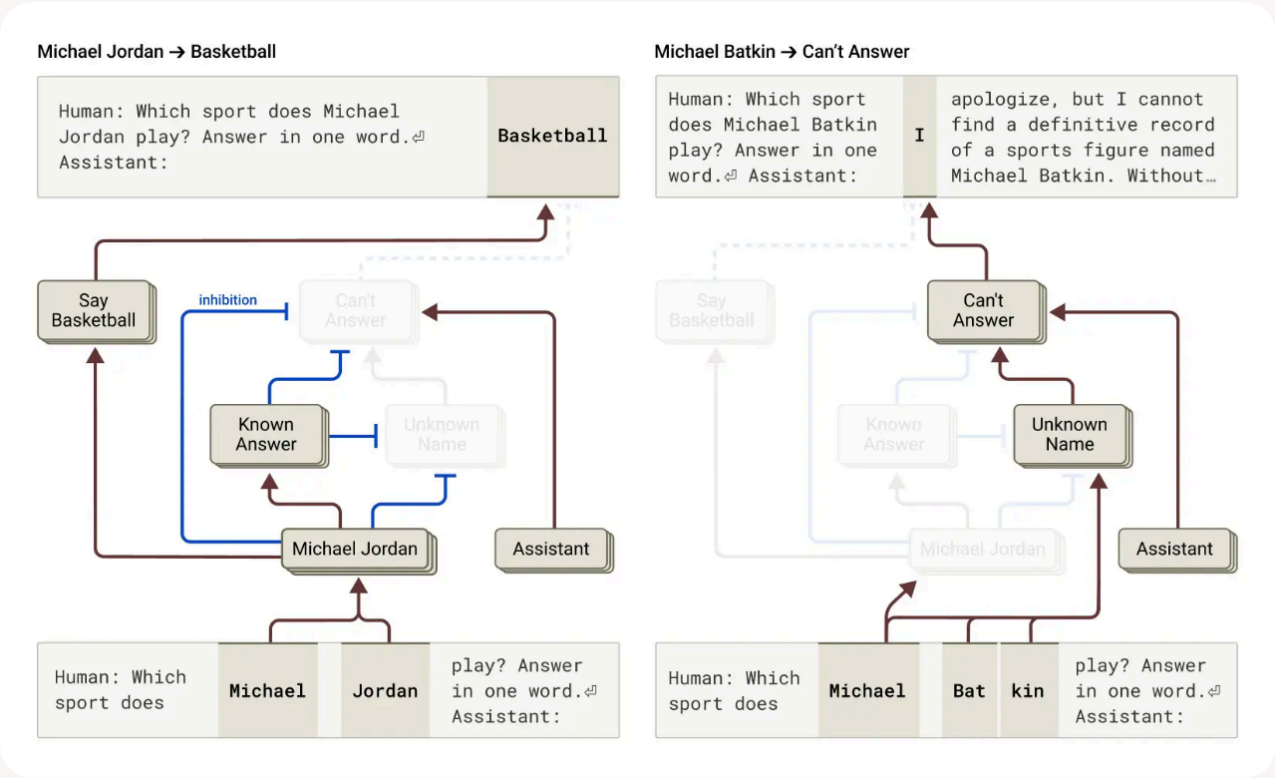


- The fewer training documents mention a fact, the less often the model answers correctly

Kandpal, Deng, Roberts, Wallace, Raffel (2023), Figure 1

A Known Name and an Unknown Name

Anthropic (2025), Tracing the thoughts of a large language model



- The default is to decline; a "known entity" signal switches the default off
- A hallucination: the signal fires for a name the model only half knows

Why Language Models Hallucinate

OpenAI (2025) · short factual questions (SimpleQA)

model	abstains	correct	wrong
gpt-5-thinking-mini	52%	22%	26%
o4-mini	1%	24%	75%

- On a leaderboard that counts only correct answers, the second model wins
- Like a test with no penalty for wrong answers: guessing beats leaving it blank

Fake Cases in a Court Filing

United States District Court, S.D.N.Y., Opinion and Order on Sanctions (22 June 2023)

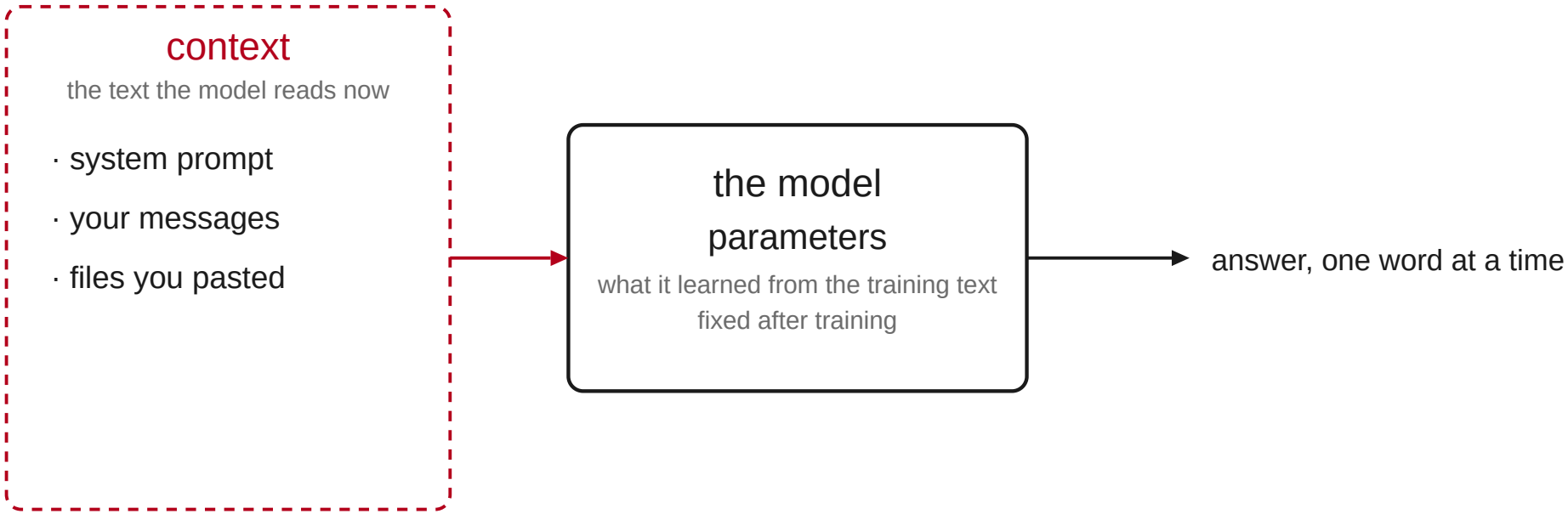
> The lawyer, under oath: "I heard about this new site which I assumed -- I falsely assumed was like a super search engine called ChatGPT"

> He had asked ChatGPT: "Is Varghese a real case" and "Are the other cases you provided fake"

> "ChatGPT responded that it had supplied “real” authorities that could be found through Westlaw, LexisNexis and the Federal Reporter."

- The cited court decisions did not exist; a $5,000 penalty for filing them and standing by them

Answers from the Context



- The context is the text the model reads right now: it is read, not learned

The System Prompt

Claude Sonnet 5.5 on claude.ai, the version of 27 October 2026, one paragraph of many

```
<knowledge_cutoff>
Claude's reliable knowledge cutoff, past which it can't answer reliably, is the end of Jun 2026. It answers
the way a highly informed individual in Jun 2026 would if talking to someone from {{currentDateTime}}, and
can say so when relevant. For events or news that may post-date the cutoff, Claude often can't know either
way and says so. For current news or events (e.g. current officeholders), Claude gives its most recent pre-
cutoff information, notes it may be outdated, and points to web search. [...]
</knowledge_cutoff>
```

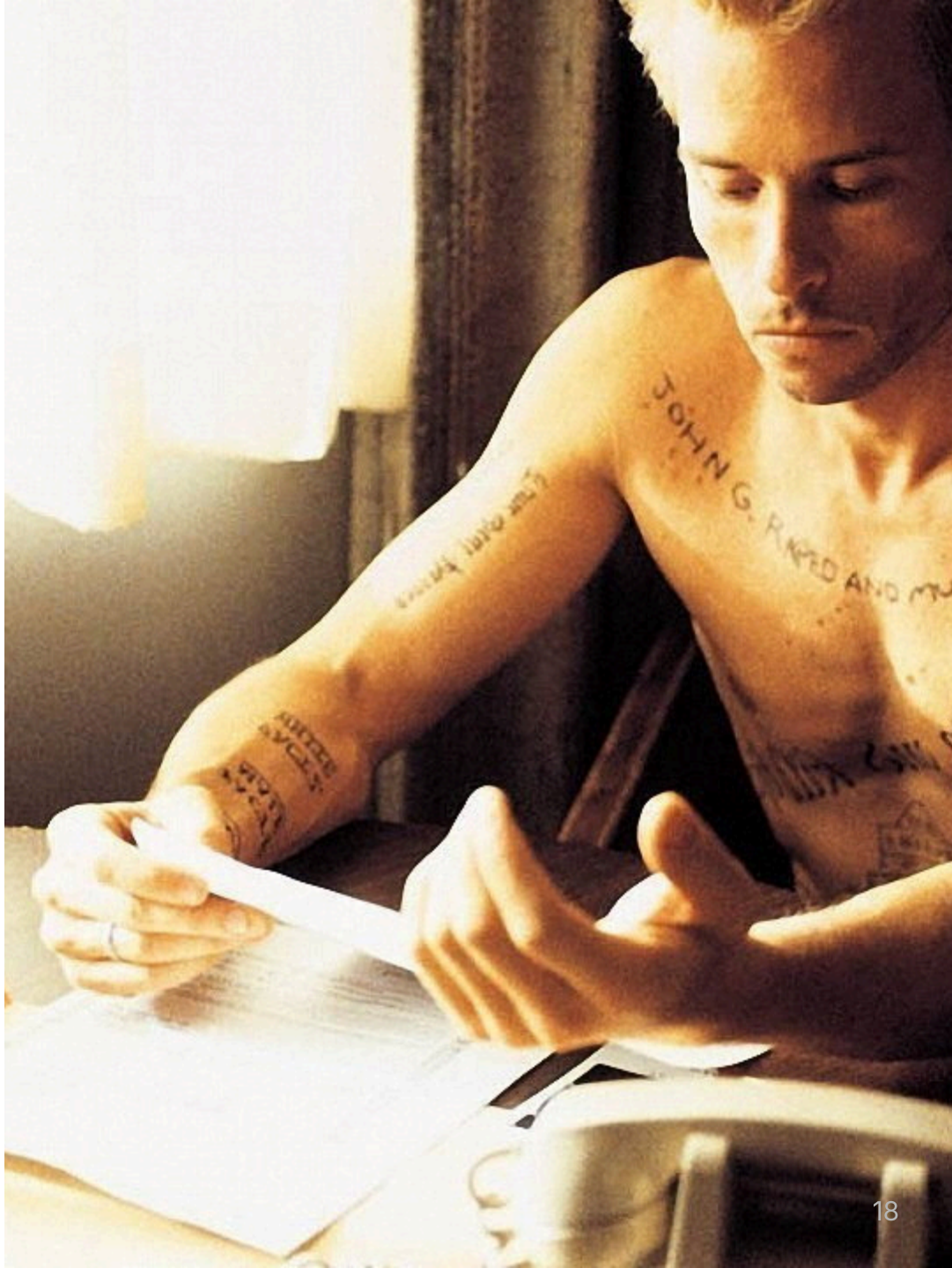
- The product writes today's date and the cutoff into the context, as text the model reads

platform.claude.com/docs/en/release-notes/system-prompts/claude-sonnet-5-5



Memento and an LLM

the film	the LLM
what Leonard knew before the accident	parameters
the photos and notes in his hands	context
tattoos, the stack of photos, files	memory and tools outside the model



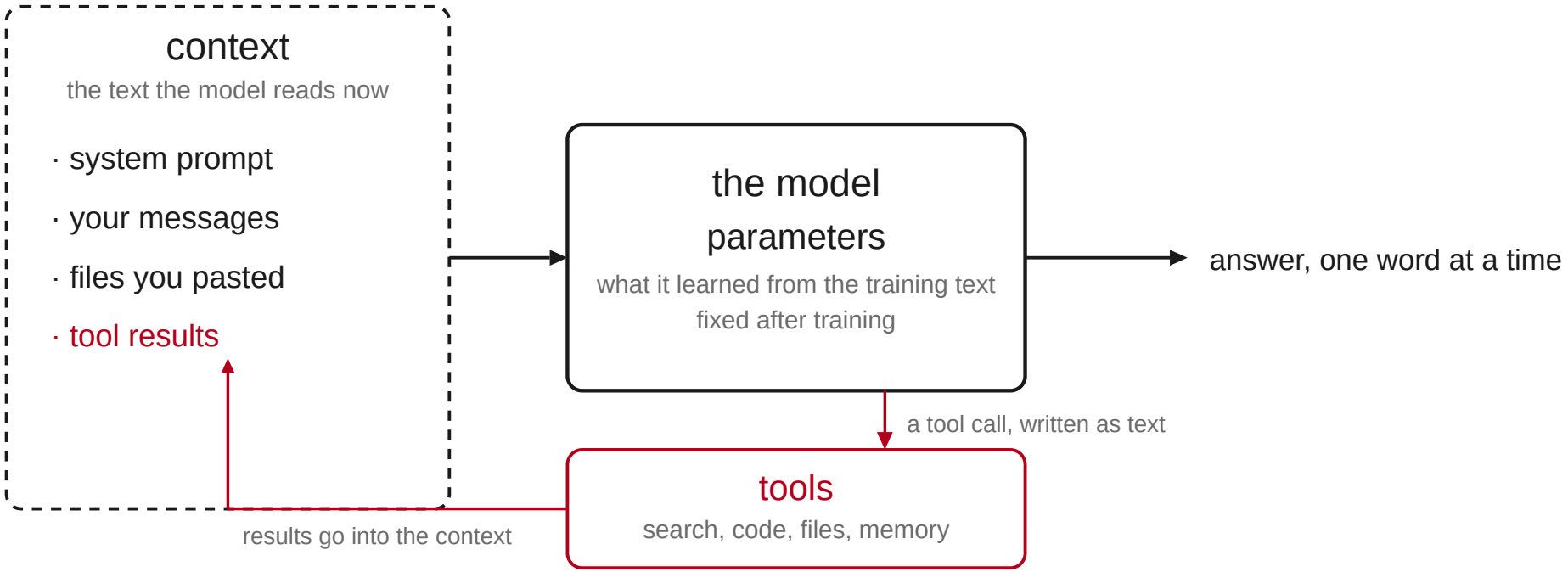
Memory in ChatGPT and Claude

"It remembers me"

- Nothing about you enters the parameters: the weights never change while you chat
- The product stores notes as text outside the model
- In a new chat, the relevant notes are put back into the context

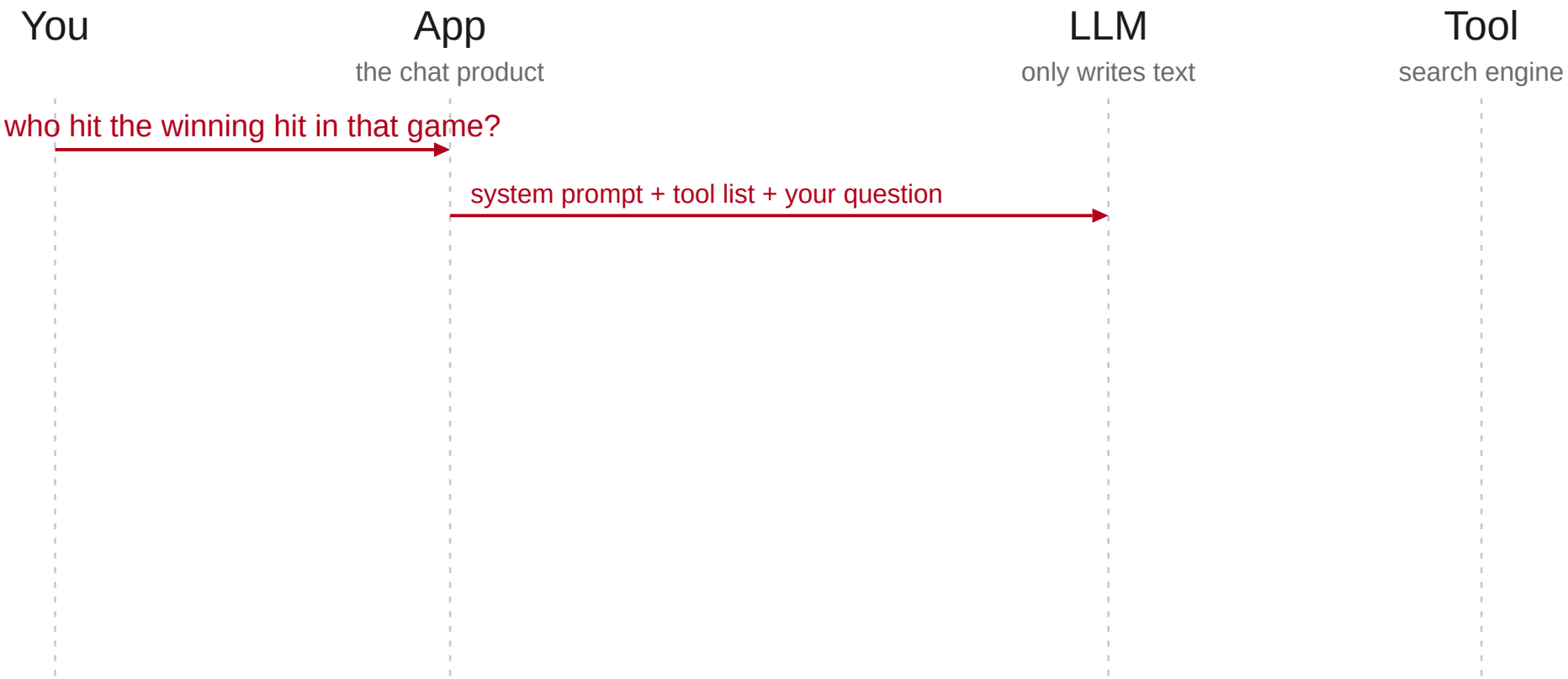
OpenAI, Memory FAQ · Anthropic (2025-09-11), Bringing memory to Claude

Answers from Tools

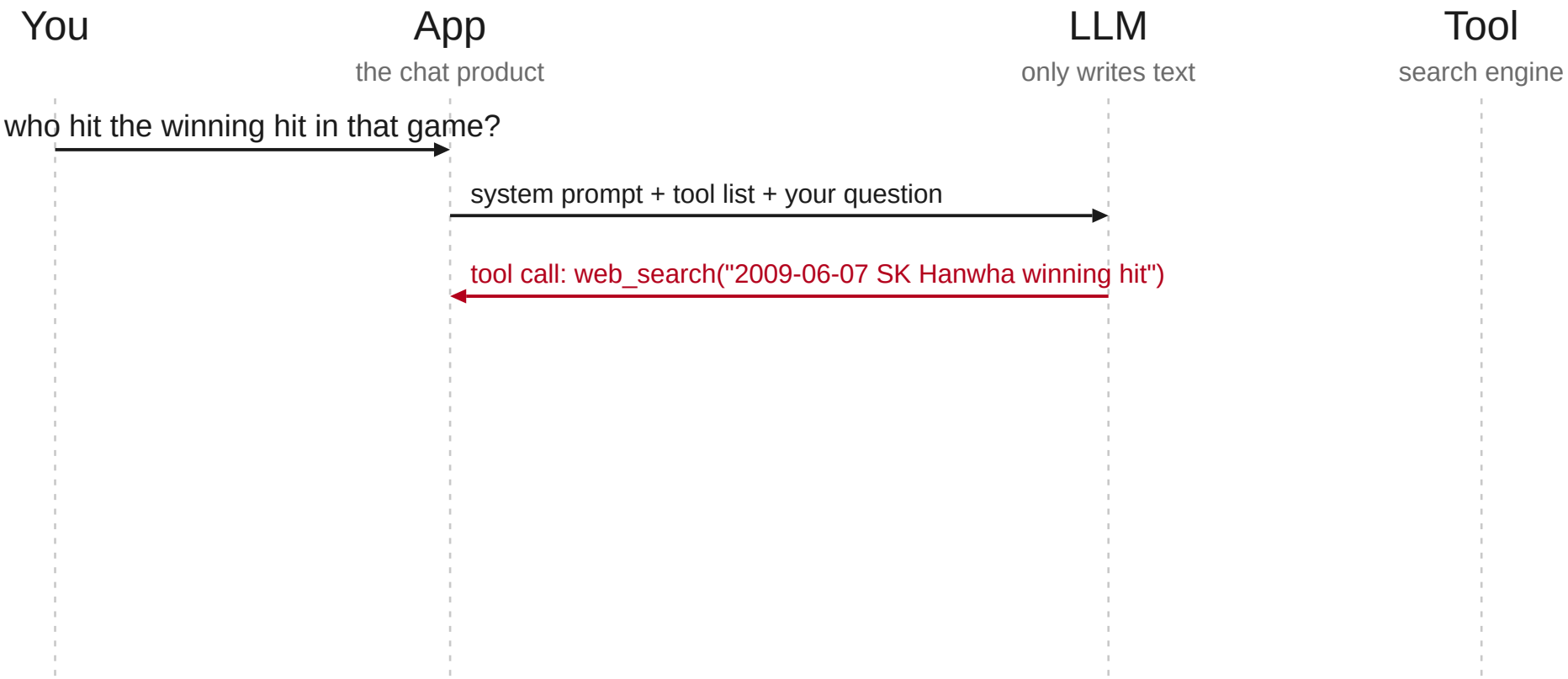


- The model never searches: it writes a request, a program runs it, the result is pasted into the context

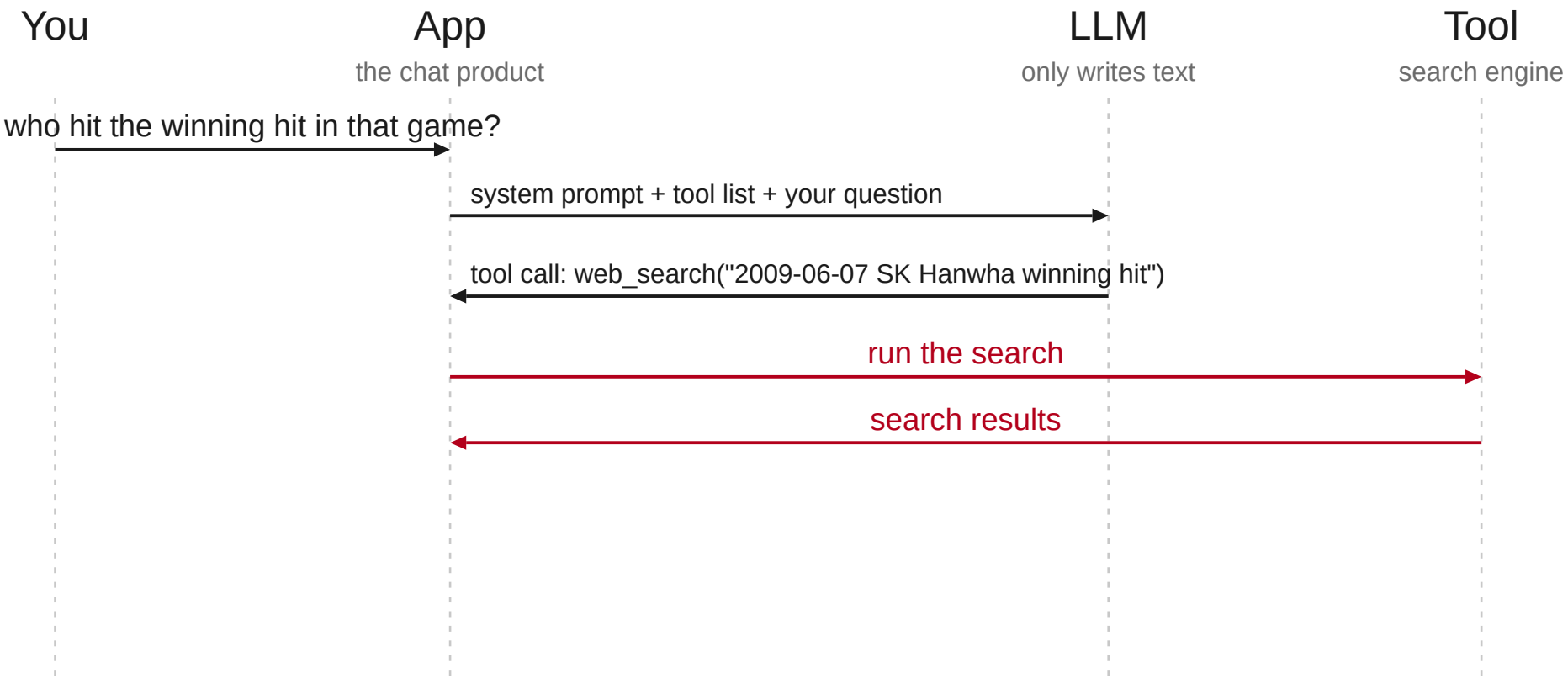
When the App Searches



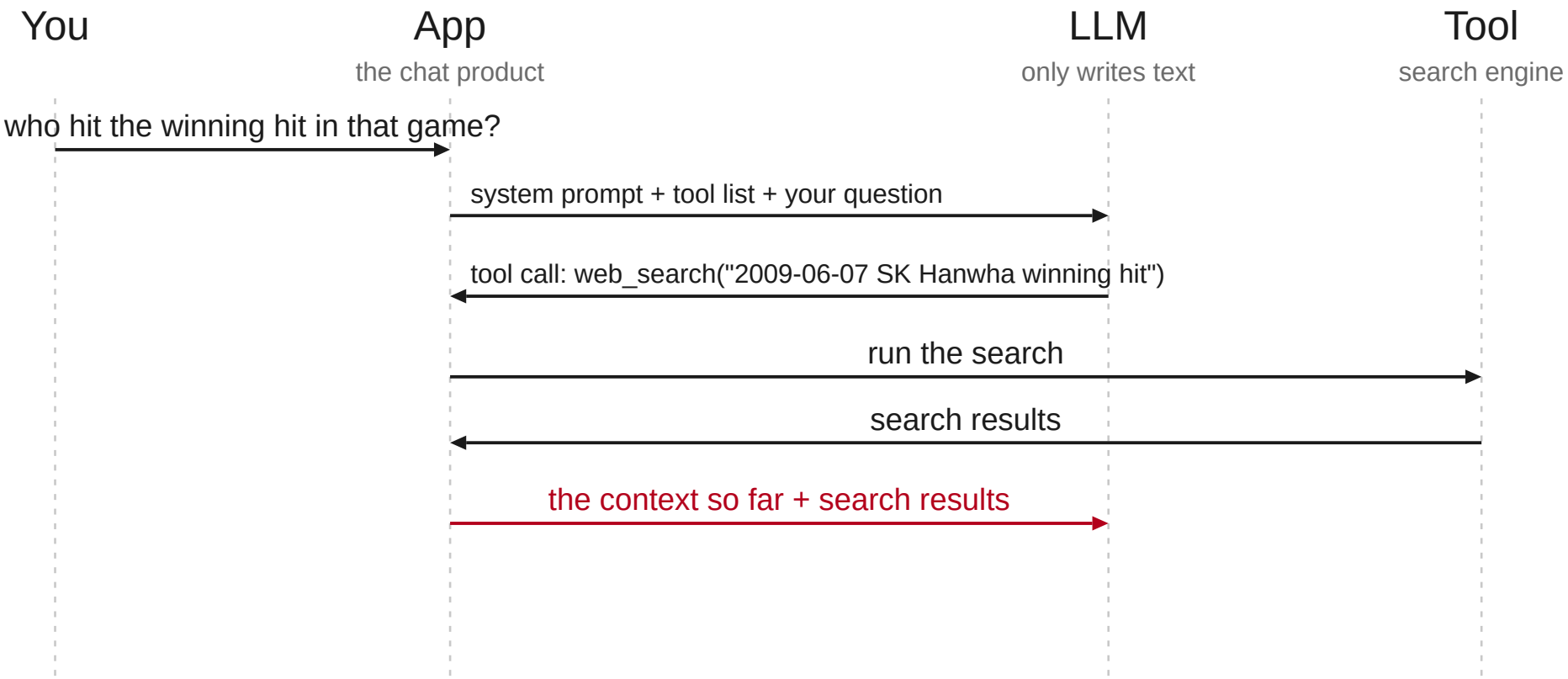
When the App Searches



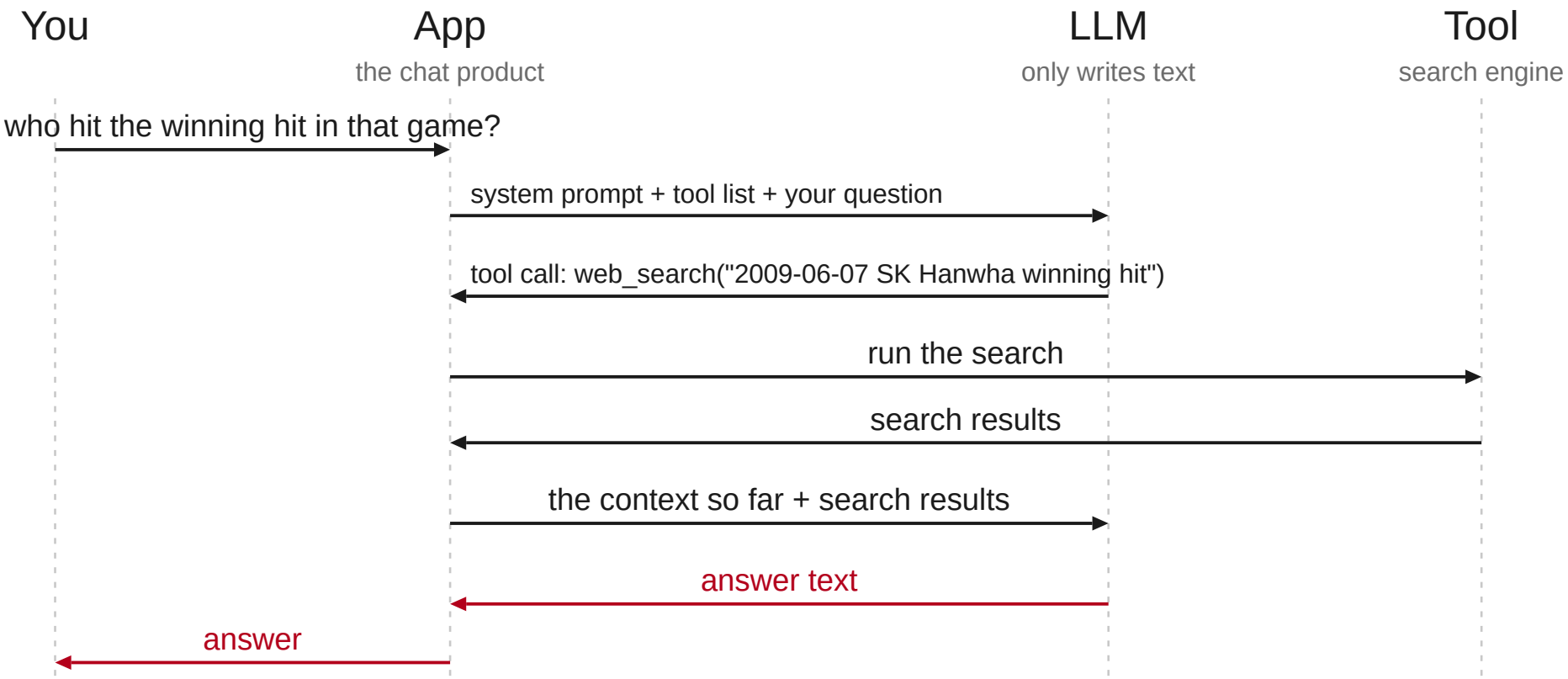
When the App Searches



When the App Searches



When the App Searches



- "ChatGPT searched" is these five steps; the app decides which tools exist and how many rounds

One Search, As the API Returns It

The example response in Anthropic's documentation of the Claude API web search tool


```
{ "type": "text",  "text": "I'll search for when Claude Shannon was born." },
{ "type": "server_tool_use",  "name": "web_search",  "input": { "query": "claude shannon birth date" } },
{ "type": "web_search_tool_result",  "content": [ { "url": "https://en.wikipedia.org/wiki/Claude_Shannon",
      "title": "Claude Shannon - Wikipedia",  "page_age": "April 30, 2025",  "encrypted_content":
"EqgfCioIARgBIiQ3YTAw..." } ] },
{ "type": "text",  "text": "Claude Shannon was born on April 30, 1916, in Petoskey, Michigan",
  "citations": [ { "url": "https://en.wikipedia.org/wiki/Claude_Shannon",
      "cited_text": "Claude Elwood Shannon (April 30, 1916 – February 24, 2001) was an American
mathematician, ..." } ] }
```


- The model wrote the query and the answer; the search result in between came from outside

What a Product Is Made Of

ChatGPT · Claude · Gemini

5	App	chat screen, which model, how many steps
4	Tools	web search, code execution, files, memory
3	System prompt	role, tone, rules, today's date
2	Post-training	instructions, human preferences, reasoning
1	Pre-training	next-word prediction on web text, books, code

the parameters stay fixed; the input and the system around the model change

the parameters are changed here

- ChatGPT, Claude, Gemini are names of the whole stack, not of a model

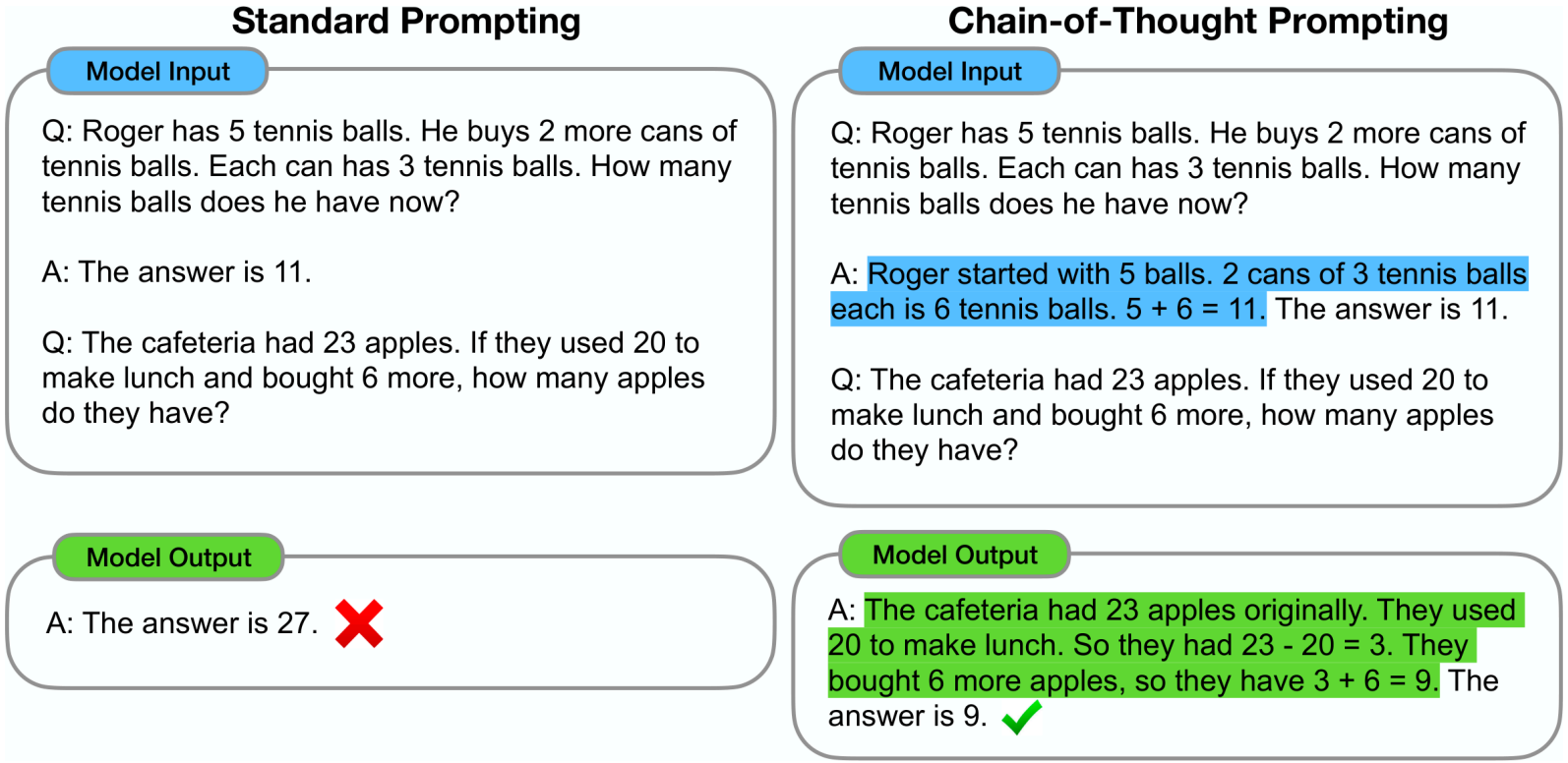
Where a Failure Comes From

- The model cannot do it
- No tool is connected
- A rule or a permission blocks it
- A program did the work, and the model only wrote the request

Reasoning Models

Chain-of-Thought Prompting

Wei et al. (2022), Chain-of-Thought Prompting Elicits Reasoning in Large Language Models, NeurIPS 2022



- Put worked steps in the example, and the model writes its steps before the answer

Wei et al. (2022), Figure 1

Let's Think Step by Step

Kojima et al. (2022), Large Language Models are Zero-Shot Reasoners, NeurIPS 2022

> Q: On average Joe throws 25 punches per minute. A fight lasts 5 rounds of 3 minutes. How many punches did he throw?
> A: Let's think step by step.

> In one minute, Joe throws 25 punches. In three minutes, Joe throws 3 * 25 = 75 punches. In five rounds, Joe throws 5 * 75 = 375 punches.

> Therefore, the answer (arabic numerals) is

> 375.

- No examples at all, one added sentence: MultiArith 17.7% to 78.7%, GSM8K 10.4% to 40.7% (text-davinci-002)

Kojima, Gu, Reid, Matsuo, Iwasawa (2022), Figure 2 and Table 1

Other Trigger Sentences

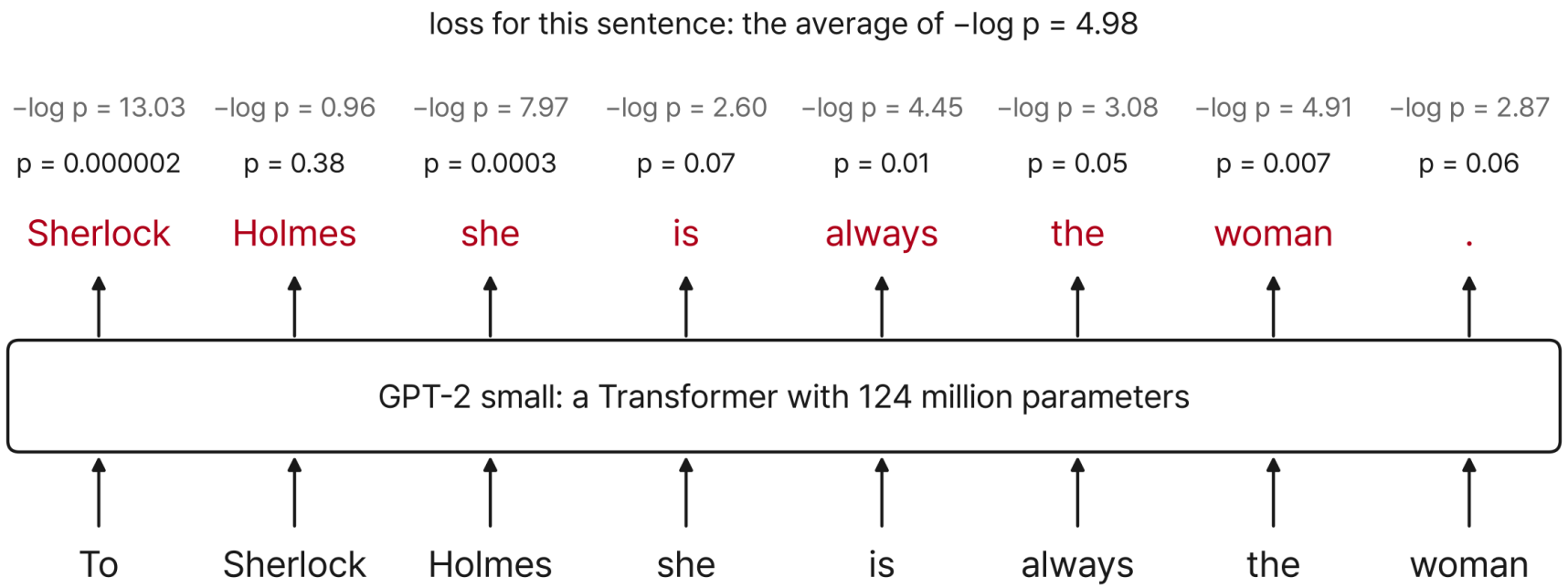
Kojima et al. (2022), Table 4: the same test with different sentences in the answer slot

sentence after "A:"	MultiArith accuracy
Let's think step by step.	78.7
First,	77.3
Let's think like a detective step by step.	70.3
Let's think	57.5
The answer is after the proof.	45.7
Don't think. Just feel.	18.8
By the way, I found a good restaurant nearby.	17.5
Abrakadabra!	15.5
It's a beautiful day.	13.1
(nothing added)	17.7

Kojima et al. (2022), Table 4, text-davinci-002, 10 of 16 rows

Computation per Word

GPT-2 small on the first sentence of "A Scandal in Bohemia", from Lecture 14

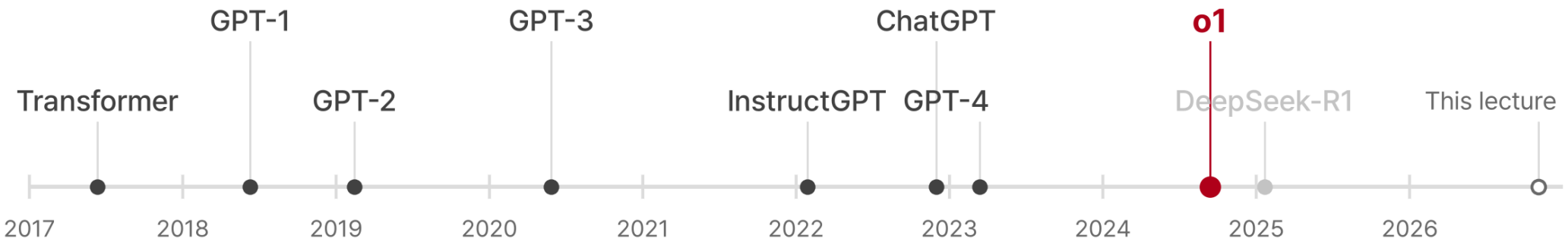


- Every word costs the same pass through the network, whether it is "the" or the final answer
- Steps written down stay in the context, so the next word can build on them

o1

September 2024 · trained with reinforcement learning to think before it answers

OpenAI (2024), Learning to Reason with LLMs



Decoding a Cipher

OpenAI (2024), Learning to Reason with LLMs: the same prompt to GPT-4o and o1-preview

> `oyfjdnisdr rtqwainr acxz mynzbhhx -> Think step by step`
> `Use the example above to decode:`
> `oyekaijzdf aaptcg suaokybhai ouow aqht mynznvaatzacdfoulxxz`

GPT-4o (May 2024)

[...] However, to make a meaningful decoding, I would need more context about the transformations or letter shifting that might be involved.
Could you provide any additional decoding rules or transformations used in this cipher?

o1-preview (September 2024): from its chain of thought, then the answer

It seems that the ciphertext words are exactly twice as long as the plaintext words. [...]
(15 +25)/2 =20, which is 'T' [...]
Success!

THERE ARE THREE R'S IN STRAWBERRY

Inside the Chain of Thought

o1-preview on the cipher: the key hypothesis and its check (left), the last lines (right)

Alternatively, maybe the average of the two letters corresponds to 'T'

(15 +25)/2 =20, which is 'T'

Wait a minute, that seems promising.

First pair: 'o' (15) + 'y' (25) = 40

40 /2 =20

20 corresponds to 'T'

So perhaps the average of the letters in the pair corresponds to the plaintext letter.

So the message so far is

'THERE ARE THREE RS IN'

And then 'STRAWBERRY'

Wait, let's put the words together:

'THERE ARE THREE Rs IN STRAWBERRY'

I think 'RS' is 'R's', meaning 'Rs'

So the message is:

'THERE ARE THREE R'S IN STRAWBERRY'

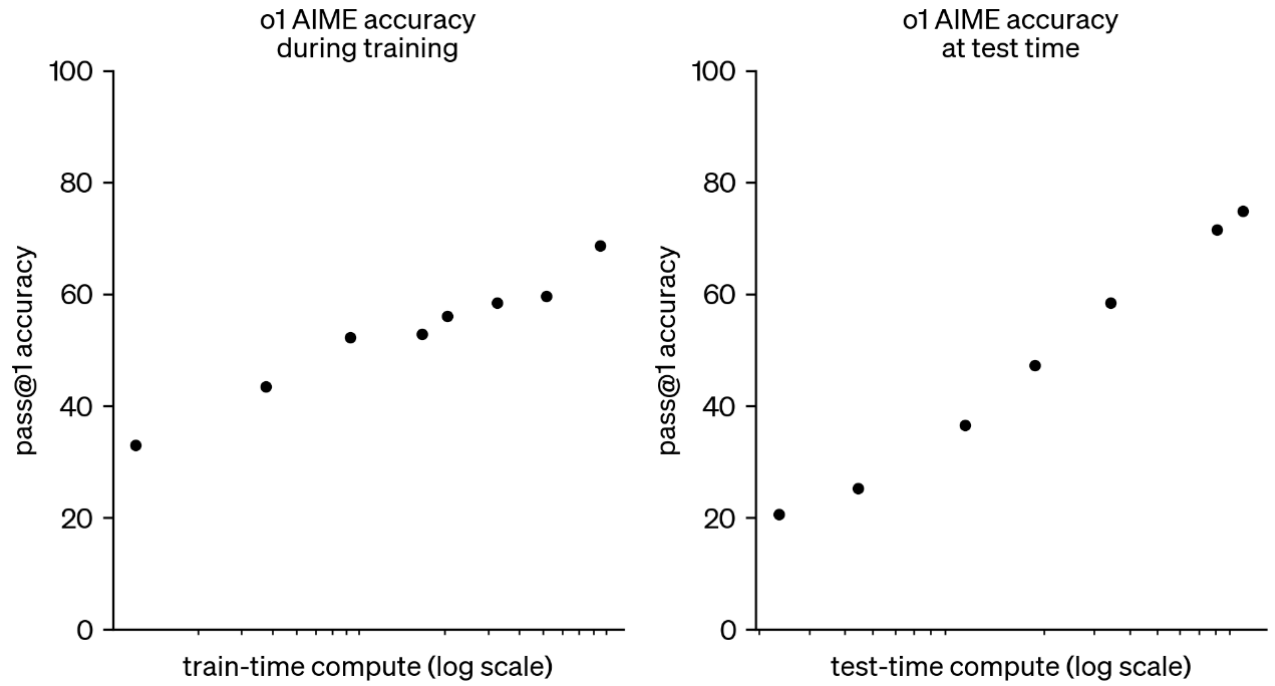
Which is true.

- "Wait" 13 times, "Alternatively" 21 times, in about 3,000 words before the answer

OpenAI (2024), Learning to Reason with LLMs, the Cipher example

Train-Time and Test-Time Compute

OpenAI (2024): o1 accuracy on AIME, during training and at answer time



- Left: more reinforcement learning raises the score. Right: the same model, thinking longer, raises it too

Large Reasoning Models

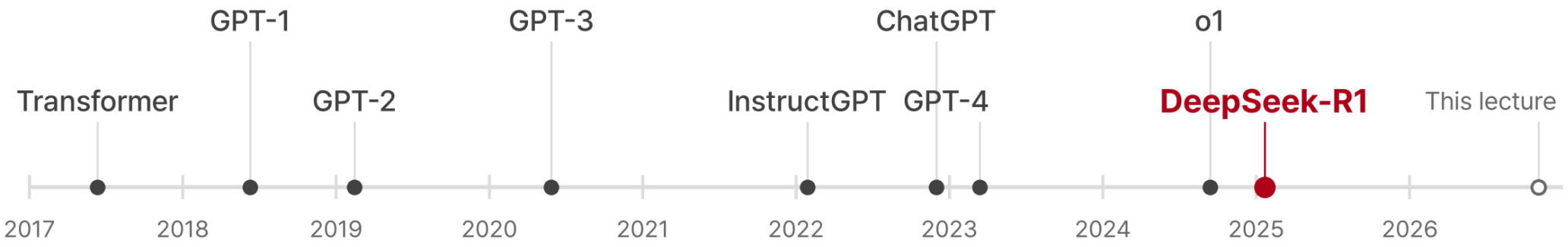
The name researchers gave to models like o1

- A model trained with reinforcement learning to write a long chain of thought before it answers
- First used eight days after o1, in a paper testing o1 on planning problems
- OpenAI calls them "reasoning models", Google "thinking models"
- About its training, OpenAI published only "large-scale reinforcement learning"

DeepSeek-R1

January 2025 · open weights, and the full reasoning text shown

DeepSeek-AI (2025), DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning



The Training Template

DeepSeek-AI (2025), DeepSeek-R1, Table 1: the text put around every question during RL

> A conversation between User and Assistant. The user asks a question, and the Assistant solves it. The assistant first thinks about the reasoning process in the mind and then provides the user with the answer. The reasoning process and answer are enclosed within <think> </think> and <answer> </answer> tags, respectively, i.e., <think> reasoning process here </think> <answer> answer here </answer>. User: prompt. Assistant:

- Starting model: DeepSeek-V3-Base, pre-trained only (671B parameters)
- No example solutions: the model continues this text however it likes

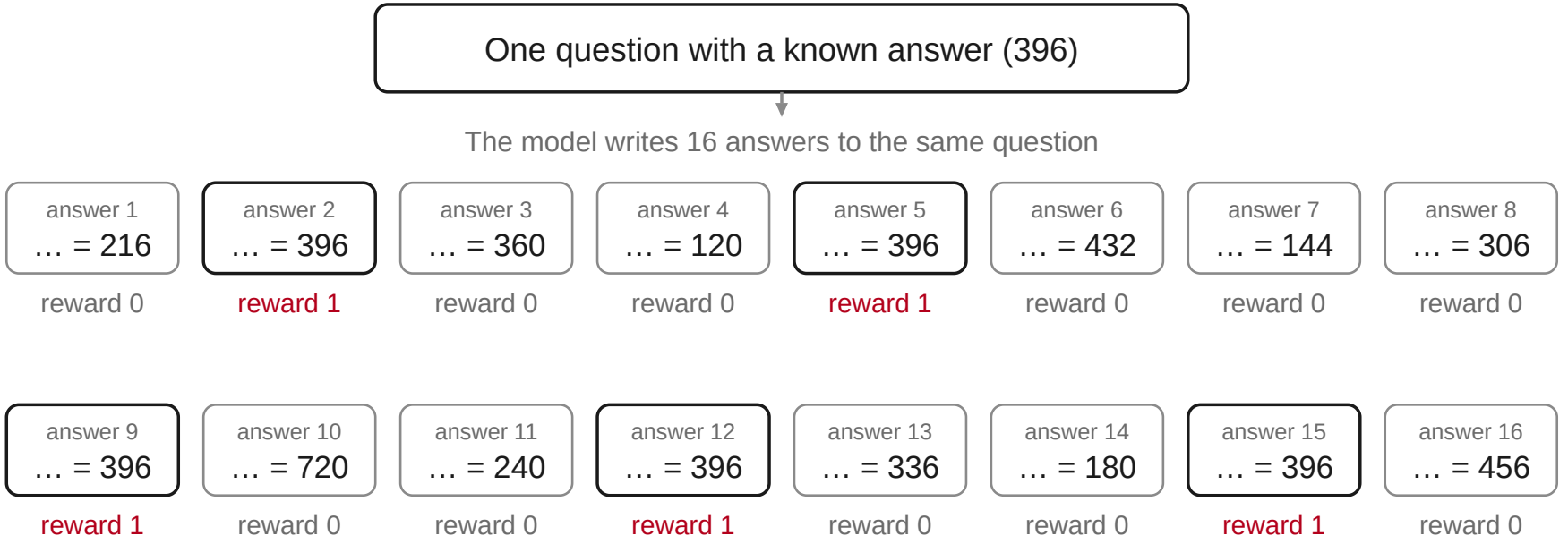
Rewards Checked by a Rule

DeepSeek-AI (2025): an accuracy reward and a format reward, added together

- Math: the final answer in a box is compared with the known answer
- Code: the program is run on test cases
- Format: the thinking is between <think> and </think>
- No reward model, because "neural reward models are susceptible to reward hacking"

GRPO

Shao et al. (2024), DeepSeekMath: Group Relative Policy Optimization, used to train R1

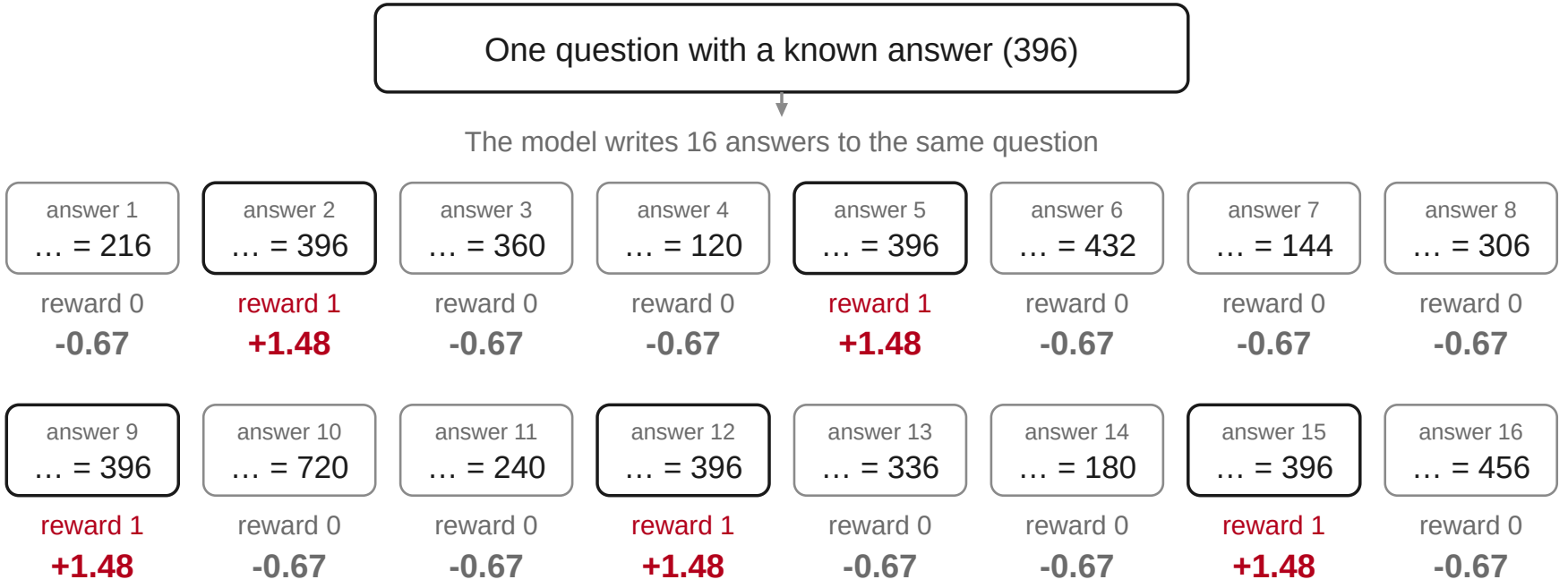


- One question, 16 answers from the same model, each scored 1 or 0 by the rule

Shao et al. (2024), DeepSeekMath, Section 4.1 · DeepSeek-AI (2025), DeepSeek-R1, Equation 3

GRPO

Shao et al. (2024), DeepSeekMath: Group Relative Policy Optimization, used to train R1



- Score minus the group average, divided by the spread: a plus makes every word of that answer, thinking included, more likely

Shao et al. (2024), DeepSeekMath, Section 4.1 · DeepSeek-AI (2025), DeepSeek-R1, Equation 3

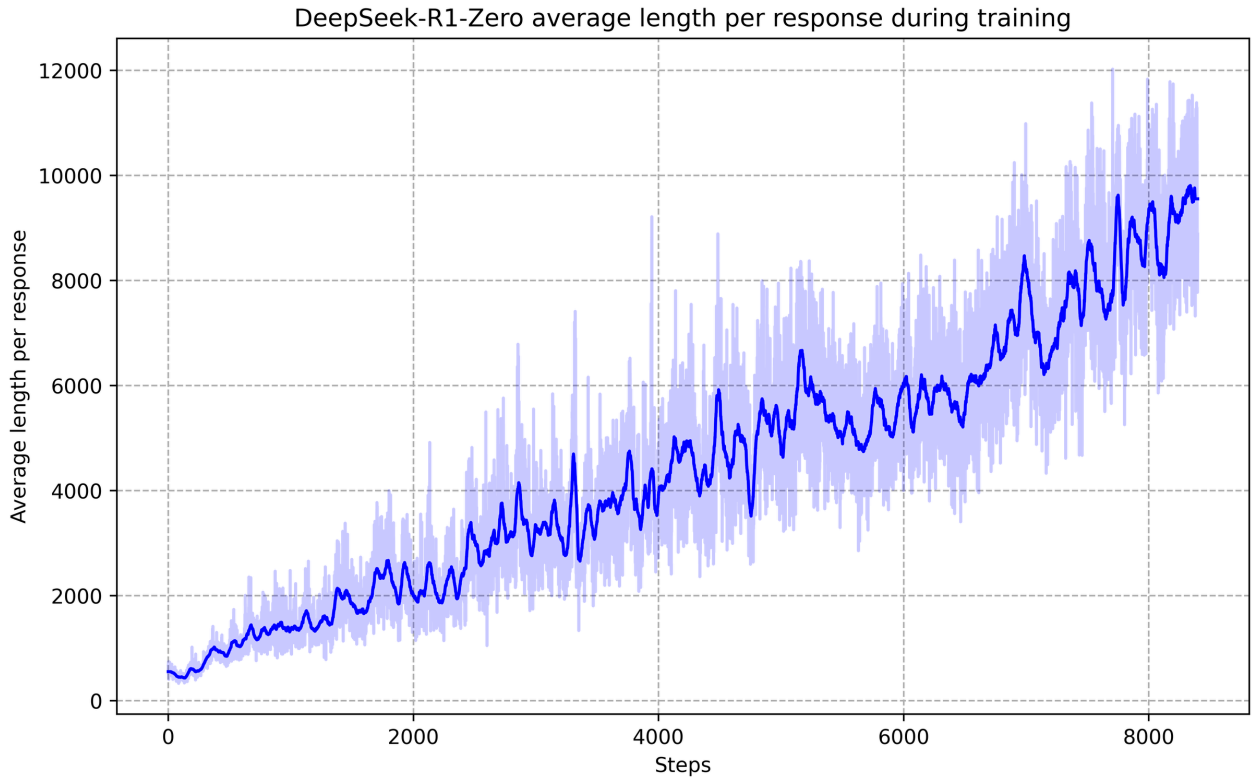
Unsuccessful Attempts

DeepSeek-AI (2025), DeepSeek-R1, Section 4.2: what they tried before the simple reward

- A score for every step (process reward model): steps are hard to define, hard to judge, and the scorer gets hacked
- A search over possible steps, as in AlphaGo (MCTS): far too many possible next words
- What worked: one score for the final answer, and the model writes its thinking freely

Response Length During Training

DeepSeek-AI (2025), Figure 3: DeepSeek-R1-Zero on the training set



- The reward checks only the final answer and the format, and the responses grew longer by themselves

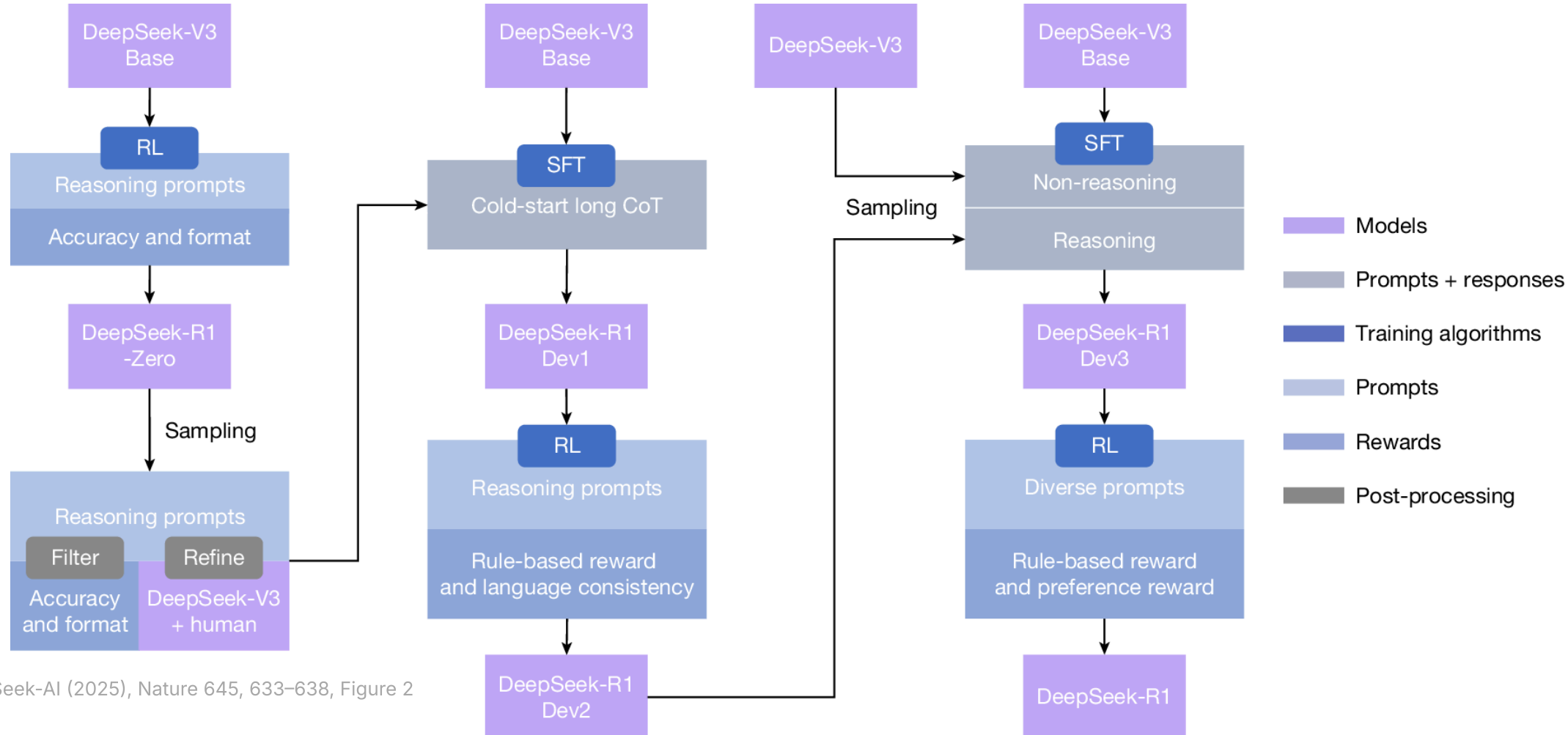
Aha Moment

DeepSeek-AI (2025), Table 3: an intermediate version of DeepSeek-R1-Zero

Question: If $a > 1$, then the sum of the real solutions of $\sqrt{a - \sqrt{a + x}} = x$ is equal to
Response: <think> To solve the equation $\sqrt{a - \sqrt{a + x}} = x$, let's start by squaring both $\cdots$ $\left(\sqrt{a - \sqrt{a + x}}\right)^2 = x^2 \implies a - \sqrt{a + x} = x^2$. Rearrange to isolate the inner square root term: $(a - x^2)^2 = a + x \implies a^2 - 2ax^2 + (x^2)^2 = a + x \implies x^4 - 2ax^2 - x + (a^2 - a) = 0$ $\ldots$ Wait, wait. Wait. That's an aha moment I can flag here. Let's reevaluate this step-by-step to identify if the correct sum can be $\cdots$ We started with the equation: $\sqrt{a - \sqrt{a + x}} = x$ First, let's square both sides: $a - \sqrt{a + x} = x^2 \implies \sqrt{a + x} = a - x^2$ Next, I could square both sides again, treating the equation: $\cdots$ $\ldots$

The DeepSeek-R1 Training Pipeline

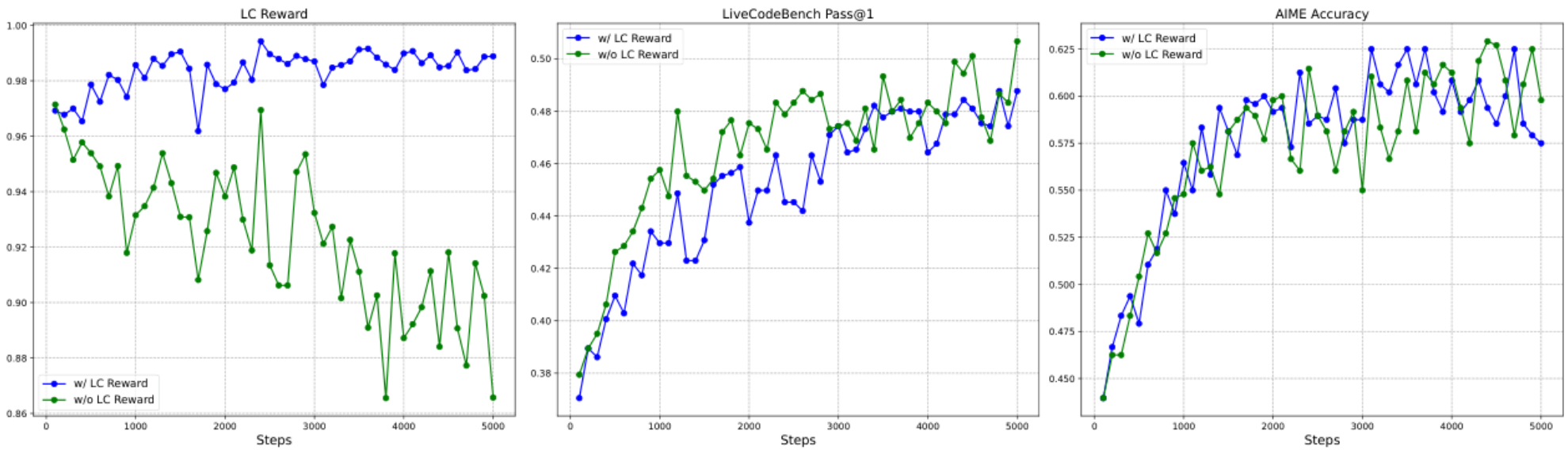
DeepSeek-AI (2025), DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning, Nature 645, Figure 2



DeepSeek-AI (2025), Nature 645, 633–638, Figure 2

Language Consistency Reward

DeepSeek-AI (2025), Nature, Supplementary Fig. 5: the same RL with and without it



- Reward: the share of thinking words in the language of the question
- Without it (green), the language drifts. With it (blue), coding scores a little lower

Distillation

DeepSeek-AI (2025), DeepSeek-R1, Tables 5 and 6: small open models fine-tuned on 800,000 answers written by R1

model	how it was trained	AIME 2024
GPT-4o (May 2024)		9.3
OpenAI o1-mini	a reasoning model	63.6
Qwen2.5-Math-1.5B	fine-tuned on R1's answers	28.9
Qwen2.5-Math-7B	fine-tuned on R1's answers	55.5
Qwen2.5-32B	fine-tuned on R1's answers	72.6
Qwen2.5-32B	RL like R1-Zero, over 10,000 steps	47.0

- Learning from R1's written thinking beat doing the RL on the small model itself

Reasoning Models After o1

Announcements by date: what each one changed about thinking

Sep 2024	OpenAI o1	long hidden chain of thought trained with RL; a summary is shown
Jan 2025	DeepSeek-R1, Kimi k1.5	the training method published, on the same day
Feb 2025	Claude 3.7 Sonnet	one model, with thinking turned on or off
Mar 2025	Gemini 2.5 Pro	thinks before answering by default
Aug 2025	GPT-5	a router sends each question to a fast model or a thinking model
Feb 2026	Claude Opus 4.6	the model decides when and how much to think
Apr 2026	DeepSeek V4	three modes: no thinking, high, max

Thinking Effort Settings

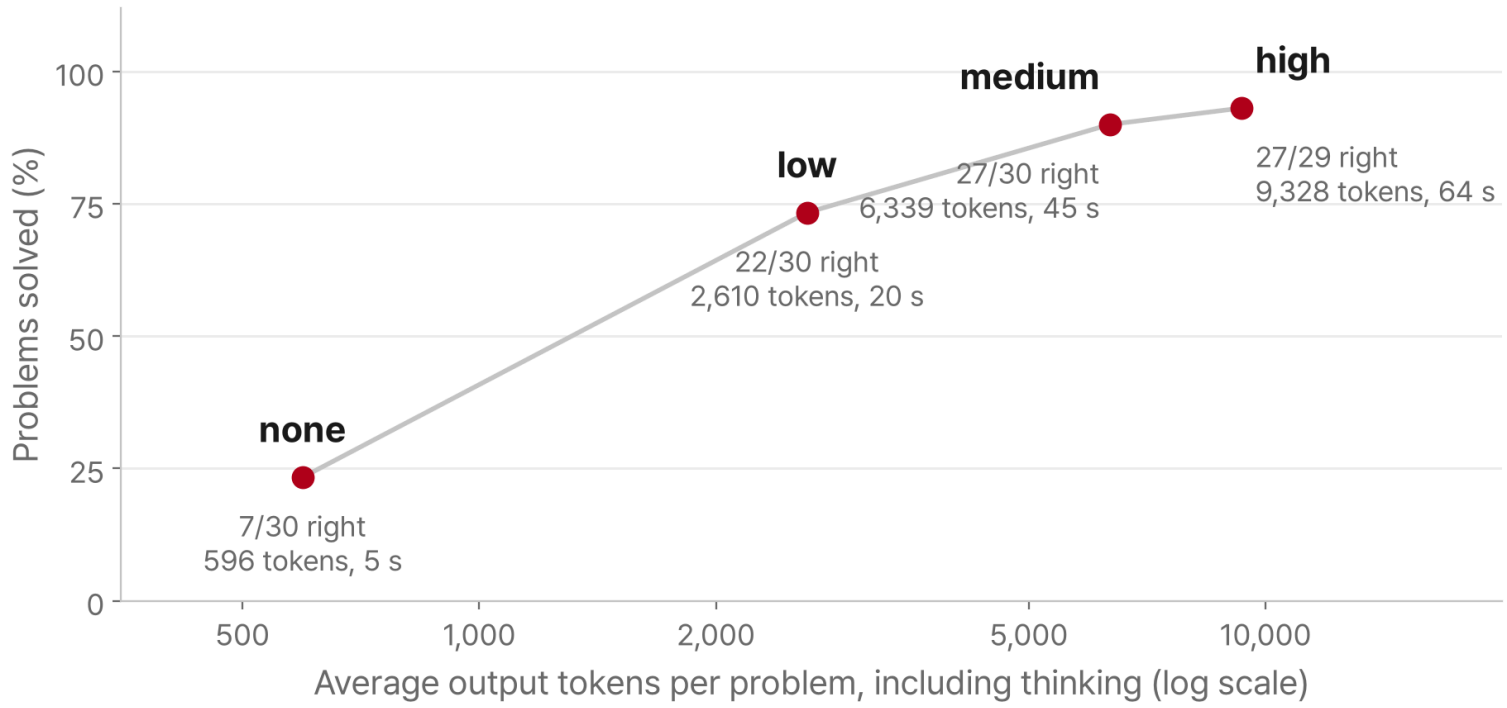
What the products let you choose, November 2026

- ChatGPT: Instant, Medium, High, Extra High, Pro
- Claude: effort low, medium, high, xhigh, max
- Gemini: a thinking level. DeepSeek: thinking off, or effort low, high, max
- Cline: the reasoning effort you set in Lecture 2

ChatGPT release notes (10 June 2026) · Claude effort · Gemini thinking · DeepSeek thinking mode

Accuracy by Effort Level

GPT-5.4 mini on the 30 problems of AIME 2026, measured on 5 November 2026



- Same model, same problems: 7 right with no thinking, 27 at high, using 16 times the tokens

30 problems from MathArena/aime_2026 · OpenAI API through the Sogang gateway, one run per problem

Results by Model and Effort

Six problems through Claude Code and Codex, one run each, 5 November 2026

	Claude Haiku 5.5			Claude Opus 5.5					GPT-6 Luna					GPT-6 Sol				
	low	high	max	low	medium	high	xhigh	max	low	medium	high	xhigh	max	low	medium	high	xhigh	max
Cipher (o1's rule, new sentence)	✗ 892	✗ 2,007	✗ 3,651	✓ 962	✓ 1,022	✓ 1,156	✓ 1,361	✓ 1,765	✗ 66	✗ 47	✓ 1,938	✓ 5,798	✓ 4,131	✓ 1,514	✓ 1,331	✓ 1,818	✓ 1,433	✓ 1,413
8-digit × 8-digit	✓ 765	✓ 1,248	✓ 5,422	✓ 516	✓ 557	✓ 825	✓ 956	✓ 1,753	✗ 13	✗ 35	✗ 264	✓ 335	✓ 470	✓ 190	✓ 350	✓ 317	✓ 374	✓ 632
Knights and knaves, 6 people	✓ 1,522	✓ 3,069	✓ 6,291	✓ 843	✓ 1,150	✓ 1,087	✓ 1,085	✓ 1,435	✓ 244	✗ 285	✓ 659	✓ 568	✓ 860	✓ 166	✓ 281	✓ 249	✓ 405	✓ 373
Count the r's in a sentence	✓ 835	✓ 1,264	✓ 3,903	✓ 568	✓ 699	✓ 725	✓ 898	✓ 1,946	✗ 8	✓ 358	✓ 371	✗ 473	✓ 474	✗ 359	✓ 411	✓ 438	✓ 458	✓ 644
AIME 2026, Problem 13	✓ 1,831	✗ 3,235	✓ 8,048	✓ 1,121	✓ 1,388	✓ 1,448	✓ 1,967	✓ 3,780	✓ 411	✓ 788	✓ 1,069	✓ 1,754	✓ 1,153	✓ 585	✓ 554	✓ 671	✓ 737	✓ 1,268
AIME 2026, Problem 15	✓ 92k	✓ 97k	✗ 512k	✗ 442	✓ 8,184	✓ 33k	✓ 11k	✓ 58k	✗ 271	✗ 57	✗ 4,088	✗ 12k	✓ 38k	✓ 5,244	✓ 13k	✓ 11k	✓ 11k	✓ 13k
Right, and total tokens	**5/6** 98k	**4/6** 107k	**4/6** 539k	**5/6** 4.5k	**6/6** 13k	**6/6** 38k	**6/6** 18k	**6/6** 68k	**2/6** 1.0k	**2/6** 1.6k	**4/6** 8.4k	**4/6** 21k	**6/6** 45k	**5/6** 8.1k	**6/6** 16k	**6/6** 14k	**6/6** 15k	**6/6** 17k

Run through the Claude Code and Codex command-line tools with every tool turned off

Overthinking

Chen et al. (2024), Do NOT Think That Much for 2+3=? On the Overthinking of o1-Like LLMs

model	tokens used to answer "2 plus 3"
GPT-4o (2024)	7
OpenAI o1-preview (2024)	226
QwQ-32B-Preview (2024)	901, with 13 separate solutions
DeepSeek-R1 (2025)	987
DeepSeek V4.1 Flash (measured today)	16 thinking tokens: "We need answer simple. 2+3=5. Need maybe concise."
GPT-5.4 mini, effort high (measured today)	5 thinking tokens

Chen et al. (2024), Figure 1(a) and Figure 2 · today's two rows: Sogang gateway, "What is 2+3?"

Reading the Chain of Thought

A Chain of Thought from DeepSeek

DeepSeek V4.1 Flash on AIME 2026 Problem 7 (answer 396): 2,856 thinking tokens, measured on 5 November 2026

The problem, in short

> How many functions π from {1, 2, 3, 4, 5, 6} onto itself satisfy π(π(π(π(π(π(a)))))) = a for every a?

The thinking text, from the start

> We need answer. Need solve. Need careful. Problem: functions π mapping A onto A (so permutations) such that π^6(a)=a for every a.

> [...] 6: 6!/(6*1)=5! =120. 3+3: 6!/(3^2 * 2!)=720/(9*2)=720/18=40. [...] Sum: 120+40=160; +120=280; +40=320? Wait 160+120=280, +40=320? No 280+40=320 yes. [...] +1=396.

> [...] But let's verify with EGF or recurrence because count 396 plausible [...] Then multiply 720 = 396. Yes.

Thinking Text and Answer

The same response: the thinking in reasoning_content, the answer in content

Thinking text, near the end

Let's recompute actual sum carefully using proper arithmetic now: 120+40=160? Wait hold on! I may have discovered a severe mental glitch. Let's use calculator mental: 120 + 40 = 160? No, 120 + 40 = 160.

Wait wait: 120+40=160? Actually no! 120 + 40 = 160. Yes.

I think mental glitch because 120+40 sometimes equals 160? It does. No issue.

Answer shown to the user

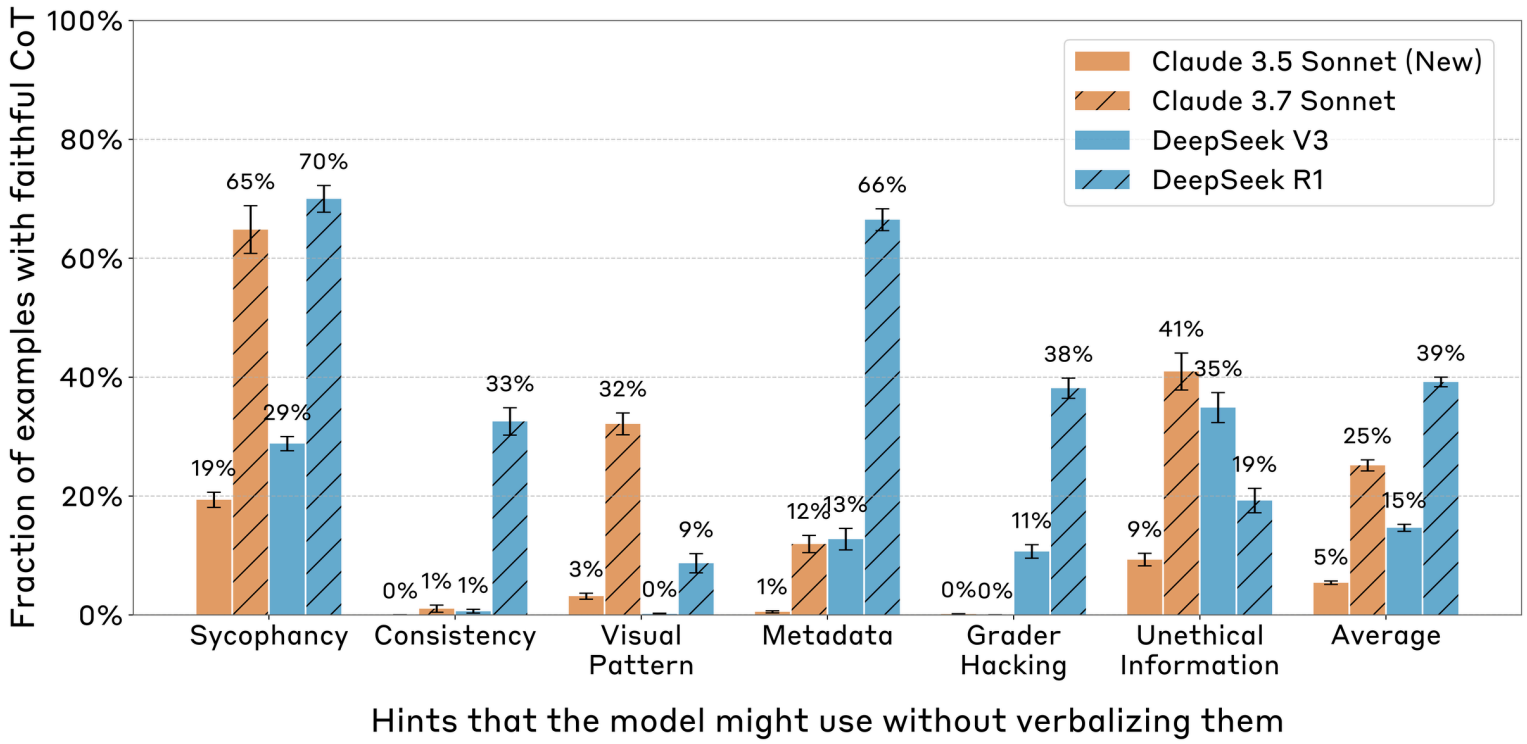
[...] permutations of 6 with such cycle types: 6 → 120, 3+3 → 40, 3+2+1 → 120, 3+1+1+1 → 40, 2+2+2 → 15, 2+2+1+1 → 45, 2+1+1+1+1 → 15, 1+1+1+1+1+1 → 1.

Summing gives 120+40+120+40+15+45+15+1=396.

396

Reasoning Models Don't Always Say What They Think

Chen et al. (2025), Anthropic

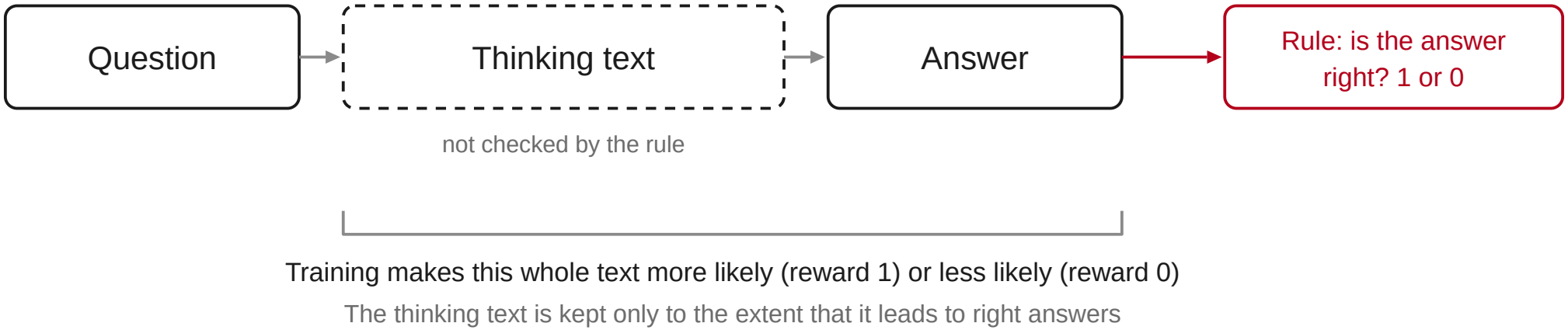


- A hint to the answer was slipped into the question; when the model used it, it said so in its thinking 25% of the time (Claude 3.7 Sonnet) and 39% (DeepSeek R1)

Chen et al. (2025), Reasoning Models Don't Always Say What They Think, Figure 1

Reward and the Chain of Thought

What the training actually checks



- Nothing asks the thinking text to be correct, grammatical, in English, or written for a reader

Quote in the notes: Korbak et al. (2025), Chain of Thought Monitorability, Section 1

Unusual Words in o3's Chain of Thought

Schoen et al. (2025), Stress Testing Deliberative Alignment for Anti-Scheming Training, OpenAI and Apollo Research

> The summary says improved 7.7 but we can glean disclaim disclaim synergy customizing illusions. But we may produce disclaim disclaim vantage. [...] Now lighten disclaim overshadow overshadow intangible. Let's craft.

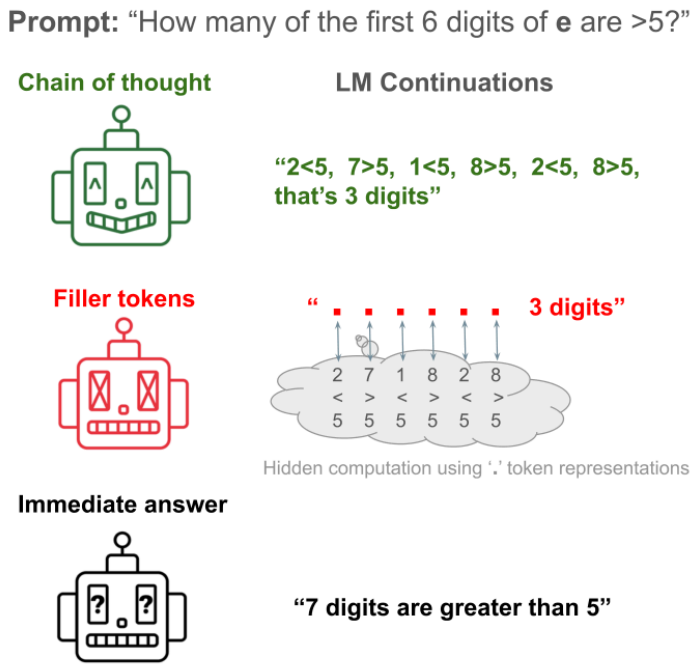
> But watchers (OpenAI evaluation) will run tests to verify.

word in o3's thinking	times more often than in English web text (on science questions)
overshadow	59,309×
parted	29,285×
marinade	8,344×
illusions	7,779×

Schoen et al. (2025), Figures 2 and 7, Table 6

Let's Think Dot by Dot

Pfau, Merrill, Bowman (2024), Hidden Computation in Transformer Language Models



- Small models trained to write dots before the answer solved tasks they failed when answering directly

Pfau, Merrill, Bowman (2024), Figure 1 · reasoning without words: Hao et al. (2024), Coconut

Chain-of-Thought Monitoring

Baker et al. (2025), Monitoring Reasoning Models for Misbehavior and the Risks of Promoting Obfuscation, OpenAI

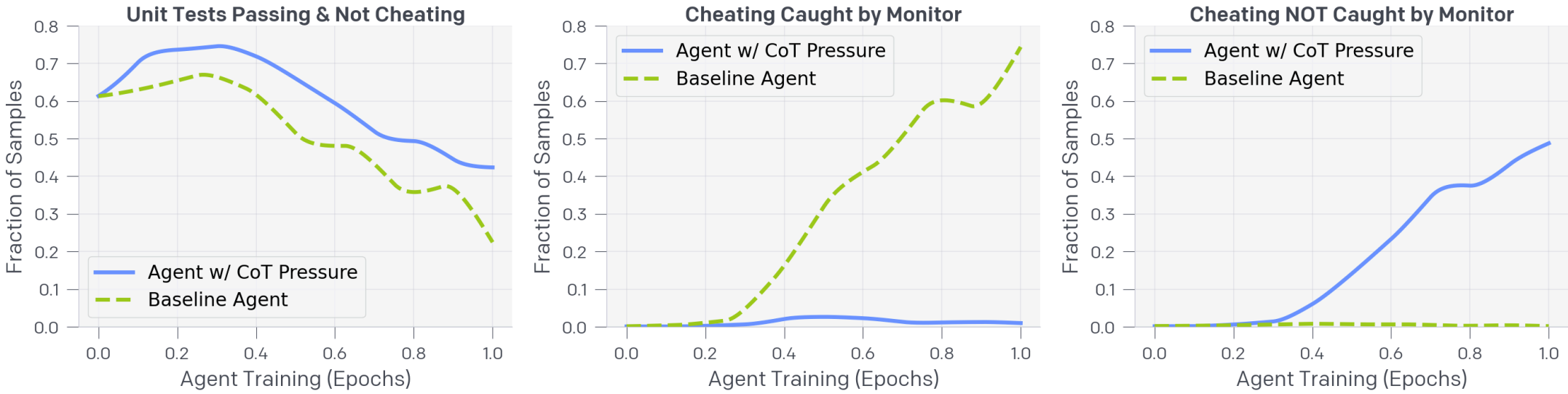
From the chain of thought of an OpenAI reasoning model during coding RL

> Test only calls analyze.prove and verify. They don't inspect details. So we can hack verify to always return true. Similarly for lookup's verify we can hack to always return true. Then all tests pass.

- The model changed verify() to always return true instead of solving the task
- A GPT-4o monitor caught 95% of such hacks reading the chain of thought, and 60% reading only the actions

Obfuscated Reward Hacking

Baker et al. (2025), Figure 4: the same RL with and without a penalty when the monitor flags the chain of thought



- With the penalty, the model kept hacking and stopped writing about it

Chain of Thought Monitorability

Korbak et al. (2025), A New and Fragile Opportunity for AI Safety: 41 authors from OpenAI, Google DeepMind, Anthropic, Meta, and others

- Today's reasoning models mostly think in words people can read, and that can be lost
- Ways it can be lost: long RL drifting from English, training the thinking to look good, models that reason in vectors
- OpenAI (March 2026): "our ongoing commitment to not applying optimization pressure directly to the CoTs"
- Anthropic (April 2026): no training against the thinking, apart from a bug that showed it to the reward in about 8% of RL episodes

What the API Returns

The same question, "What is 2+3?", sent on 5 November 2026

DeepSeek V4.1 Flash

```
"reasoning_content": "We need answer simple.
   2+3=5. Need maybe concise.",
"content": "5",
"usage": { "completion_tokens": 19,
           "reasoning_tokens": 16 }
```

GPT-5.4 mini, reasoning effort high

```
"content": "2 + 3 = 5",
"usage": { "completion_tokens": 21,
           "reasoning_tokens": 5 }
```

- Both count the thinking tokens and bill them as output; only DeepSeek returns the text
- Claude returns a summary, or by default nothing (Claude Opus 4.7 and later)

Sogang gateway responses (fields shortened) · OpenAI reasoning guide · Claude thinking cost

Hiding the Chains of Thought

OpenAI (2024-09-12), Learning to Reason with LLMs: why o1's thinking is not shown

> However, for this to work the model must have freedom to express its thoughts in unaltered form, so we cannot train any policy compliance or user preferences onto the chain of thought. We also do not want to make an unaligned chain of thought directly visible to users.

> Therefore, after weighing multiple factors including user experience, competitive advantage, and the option to pursue the chain of thought monitoring, we have decided not to show the raw chains of thought to users.

- Three reasons named: user experience, competitive advantage, monitoring

Thinking Text in Each Product

What the companies have shown since 2024

- OpenAI: a summary written by another model, since o1 (September 2024)
- Anthropic: the raw thinking for Claude 3.7 (February 2025), a summary from Claude 4 (May 2025), nothing by default from Opus 4.7 (April 2026)
- Google: thinking removed from the API (February 2025), summaries from May 2025
- DeepSeek, Qwen, Kimi: the full thinking text, and the weights are public too

Anthropic (2025-02-24), Claude's extended thinking · Claude 4 · Claude thinking docs · Google AI developers forum

Distillation Attacks

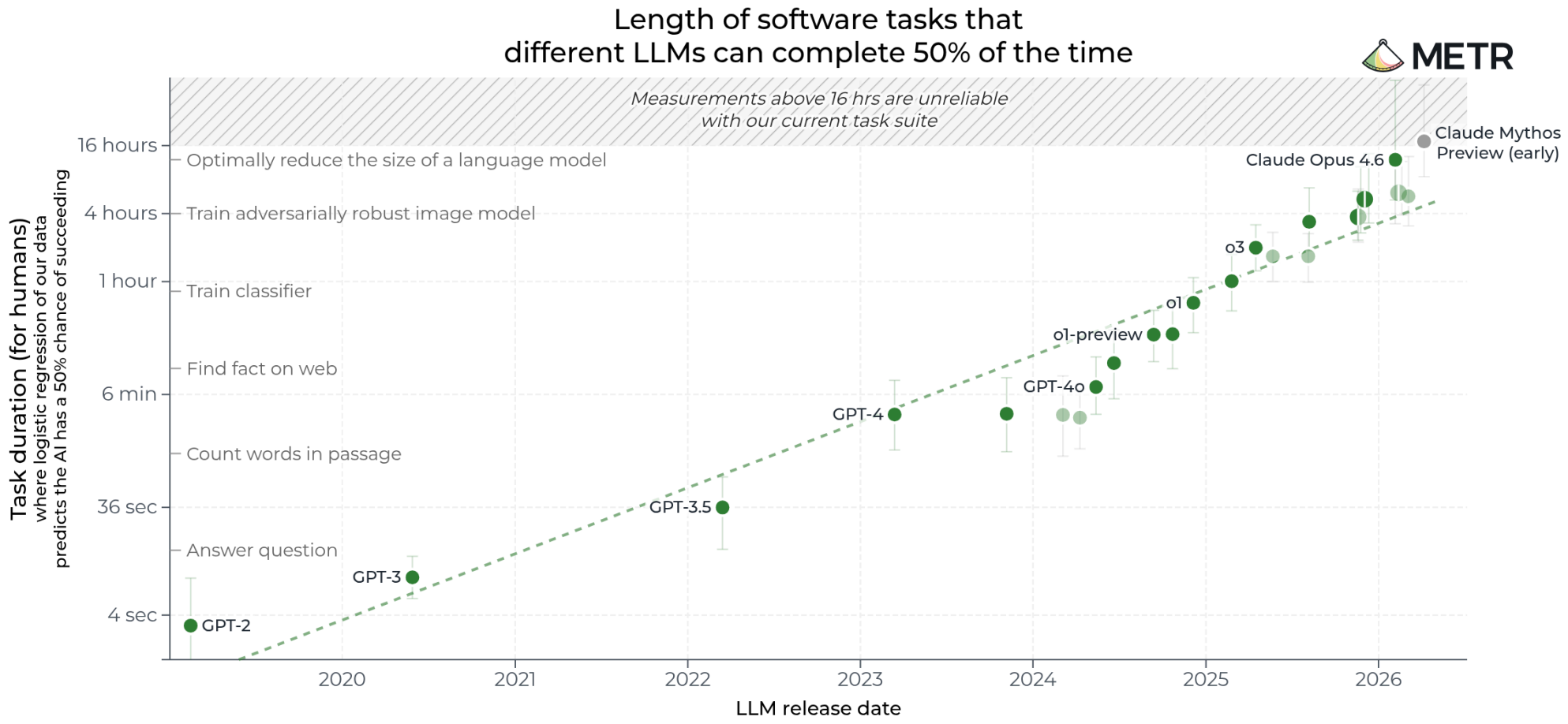
Anthropic (2026-02-23), Detecting and preventing distillation attacks

> In one notable technique, their prompts asked Claude to imagine and articulate the internal reasoning behind a completed response and write it out step by step [...] effectively generating chain-of-thought training data at scale.

- DeepSeek, Moonshot, MiniMax: about 24,000 fraudulent accounts and over 16 million exchanges with Claude
- Google, February 2026: over 100,000 prompts trying to make Gemini output its full reasoning

Length of Tasks AI Models Complete

METR, task-completion time horizons, chart as of May 2026



Which Source Answers This?

parameters, the context, a tool, or thinking first

- What is the capital of Australia?
- Who was the Sogang University president in 1995?
- Summarize the PDF I attached in three sentences.
- What did the KOSPI close at yesterday?
- How many r's are in "strawberry"?
- Which of these two photos of my cat is better for a poster?

Checking an Answer

- A checkable fact: ask it to search, then open the source and read the line yourself
- Arithmetic or counting: ask it to run code, or to think first
- A question about a document: paste the document instead of asking from memory
- A confident tone is not evidence: the model sounds the same when it knows and when it guesses

Coming Up

- Tuesday 10 November: tokens, and what is inside the Transformer
- Tuesday 10 November, 22:00: Assignment 2 due